CS 411 – Software Engineering
Term 1 – 2023/2024
Project Proposal Form

- **Group :** C2
- **Members of the group:**

#	Name	ID	Role
1			Leader
2			Member
3			Member
4			Member

- **Title of the Project:**

EDUassist

- **Introduction of the Project**

Embark on a journey where academic excellence meets technological innovation with our groundbreaking AI-powered assistant. Imagine a world where your academic routine becomes a seamless experience, effortlessly organized and optimized by cutting-edge tools and methods. Our project is designed to revolutionize the way students engage with their studies, offering a myriad of features that cater to every facet of the learning process. From transforming complex syllabi into manageable study plans to fostering collaborative brainstorming sessions on a virtual whiteboard, every aspect is meticulously crafted to enhance your educational journey. With the power of AI, visual learning is redefined through animated lectures, and resource suggestions tailored to your needs are just a click away. Embrace the future of education with our AI assistant, where peer collaboration and personalized study strategies converge to elevate your academic success to new heights.

- **Problem Statement:**

in the realm of academia, students often face challenges in efficiently managing their academic responsibilities and optimizing their learning experience. Traditional study planning methods lack the flexibility and personalized approach needed to cater to individual student needs, resulting in a lack of structure and organization in their educational journey. The absence of seamless integration between academic tools and resources further complicates the process, hindering students from accessing a unified platform for effective scheduling and collaboration. Additionally, the limited availability of interactive and visual learning aids impedes students' comprehension and engagement with course material, leading to suboptimal learning outcomes. These obstacles underscore the urgent need for an innovative solution that harnesses the power of AI technology to revolutionize academic routines and empower students to navigate their educational path with efficiency, collaboration, and enhanced learning capabilities.

- **Objectives (Product Characteristics and Requirements):**

Our AI-powered assistant project is designed with specific product characteristics and requirements in mind to ensure its effectiveness and usability in enhancing the academic experience. The key objectives of the project include:

1. Study Plan Organization: The system must be able to accurately interpret and organize syllabus information from PDF files into a user-friendly weekly schedule format for easy student tracking and time management.
2. App Integration: Seamless integration with external tools, such as Google Calendar, is essential to ensure smooth scheduling and coordination of academic tasks across platforms.
3. Collaborative Whiteboard Functionality: The system should provide a user-friendly and interactive shared whiteboard feature that enables students to engage in collaborative brainstorming and mind mapping activities with AI assistance for idea generation and error correction.
4. Visual Learning Tools: The AI technology must be capable of converting lecture text into visually engaging animated content, such as cartoons, to facilitate better understanding and retention of academic material.
5. Resource Recommendations: The system should utilize AI algorithms to analyze students' academic needs and preferences, providing personalized recommendations for study materials like videos, articles, and books relevant to their subjects.
6. Peer Collaboration Platform: The project must incorporate features that enable students to connect with their peers for knowledge sharing and collaborative learning experiences through interactive

tools like the smart whiteboard and video calls.

By meeting these objectives and product requirements, our AI-powered assistant will offer a comprehensive solution to streamline academic routines, promote interactive learning experiences, and facilitate effective collaboration among students, ultimately enhancing their academic success and overall learning outcomes.

- **Project end user:**

  Students who are eager to enhance their academic journey, optimize their study routines, collaborate with peers, and leverage cutting-edge technology to boost their learning potential will find our AI-powered assistant to be a valuable companion in their educational endeavors.

- **Summary of Project Deliverables:**

  The EDUassnt application aims to transform the academic landscape by offering a comprehensive range of features designed to enhance student learning. It includes automatic study plan organization based on uploaded syllabi, seamless integration with tools like Google Calendar for task management, and a collaborative whiteboard for brainstorming with AI support. Additionally, the app provides visual learning tools that convert lecture text into animated content, personalized study material recommendations, and a peer collaboration platform for knowledge sharing and interactive learning. These deliverables streamline academic routines, foster engagement, and promote collaborative success.

- **Project Success Criteria:**

  - The AI-powered assistant should have a user-friendly interface that is intuitive and easy to navigate for students.
  - All project deliverables, including study plan organization, app integration, collaborative whiteboard, visual learning tools, resource recommendations, and peer collaboration platform, must be completed within the agreed-upon timeline.
  - The development cost of the software should remain within the defined budget to ensure financial feasibility and efficiency.
  - The project must cater to the needs of all end users, particularly students, and ensure their satisfaction by providing a comprehensive and effective solution for enhancing their academic experience.

- **Solution Methodology:**

  We will adopt an Agile process with an Incremental model for development. This approach allows us to build the application in small, manageable increments, each adding specific features based on user feedback. Initially, we will gather requirements through surveys and interviews, followed by the creation of prototypes for early testing. Each increment will focus on implementing and refining particular functionalities, ensuring that the final product effectively meets user needs. This methodology streamlines academic routines and empowers students to achieve their educational goals.

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Approved Not approved Date: …….

Comments: